Musab Ahmed Khan

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Education

Sabanci University | **B.Sc. Computer Science and Engineering** | Istanbul, Turkey *Sep 2022 - Jun 2026*
CGPA: 3.89/4.0 (High Honors) | **100%** Merit Scholarship (Sakip Sabanci)
Key Coursework: Data Science, Spatial Data Science, Machine Learning, Deep Learning, LLMs, Artificial Intelligence, NLP, Network Science, Data Analytics and Optimization, Software Engineering, Automated Debugging, Mobile Application Development, Data Structures and Algorithms, Database Systems, Linear Algebra, Discrete Math, Probability, Statistics
Volunteering: TOM@University, SUBUH Super Active member

Bal Bhawan School | **High School** | Bhopal, India *Jun 2021*
Grade: 90%
Key Coursework: Physics, Chemistry, Maths, Computer Science

Research Experience

Teknofest 2026 - AI in Healthcare: DeepMyelinAI - MS Clinical Decision Support System *Mar 2026 - Present*

- Developing an Explanable-AI based Clinical Decision Support System (**CDSS**) for Teknofest 2026 for early diagnosis, disease subtype classification, progression prediction and relapse prediction in Multiple Sclerosis (**MS**) using CSF biomarkers, clinical disability scores, genetic risk profiles, longitudinal symptom histories, and demographic data.
- Designing a multi-stage ML framework targeting the four clinical objectives, employing specialized model types at different stages while ensuring complete transparency for clinical use with **xAI** methods like SHAP, LIME and attention visualization.

Anatomy-Aware Multi-Stage Medical Image Segmentation *Oct 2025 - Feb 2026*

- Developed medical image segmentation **CV** models for **lung** pneumothorax (SIIM-ACR) and **brain** MRI FLAIR (LGG segmentation), achieving Dice **0.85** and **0.92** respectively. Implemented a perfect accuracy routing classifier to dispatch images to these models.
- Implemented UNet++ architecture with EfficientNet (B1 for lung and B3 for brain) encoders using hybrid BCE+Dice loss and **SCSE attention** modules, with class rebalancing, threshold optimization and test-time augmentation (**TTA**) for robust inference and maxpool downsampling to preserve small pathological regions.
- Extended pipeline to include modular post-training reliability flagging mechanism combining MC Dropout uncertainty, domain-shift detection and morphological plausibility and compared performance and robustness against heterogeneous models.
- Future plans to add more model types and integrate **xAI** methods (attention maps, saliency) to support clinical decision-making.
- **Github**: https://github.com/orgs/CS415-MultiModal-Med-Image-Segmentation/repositories

SemEval 2026 Task 5: Ambistory Word Sense Plausibility Prediction *Oct 2025 - Feb 2026*

- Built a transformer-based **NLP** model for SemEval 2026 Task 5 to predict human plausibility ratings (1–5) for word senses in ambiguous narrative contexts and achieved **0.73** combined score and rank **14** on the SemEval leaderboard using fine-tuned **DeBERTa** and novel hybrid CORAL + MSE hybrid loss with dynamic weighting.
- Future plans include using LLMs to generate augmented datasets in Ambistory's style to improve generalization.
- **Github:** https://github.com/NLP-445-SemEval/SU-NLP-29

Graduation Project: AI Research Gap Finder *Feb 2025 - Jan 2026*

- Developed an **AI**-powered end to end human-in-the-loop (**HITL**) framework to automate Systematic Mapping Studies (**SMS**) using **LLMs** and Model Context Protocol (**MCP**) agents.
- Made an end to end pipeline of automating months-long SMS process of research paper retrieval, screening with snowballing, topic extraction and classification, knowledge graph construction for research trend and gap analysis, and report generation.

The Right Word at the Right Time: Adaptive Turkish Vocabulary Recommender *Feb 2026 - June 2026*

- Built a terminal-based Turkish vocabulary recommender for learners using FAISS + multilingual **bi-encoder** + weighted feature reranker (semantic fit, corpus frequency, Turkish morphological overlap, MMR diversity, difficulty penalty) with optional cross-encoder and OpenRouter LLM tutor to suggest semantically appropriate next words from a known-word learner profile.
- Best config (mpnet + mean embedding + weighted feature) reached a combined score of **0.671** (**+12.3%** vs bi-encoder-only).
- **Github**: https://github.com/MAKhan22/symmetrical-broccoli

TraceLens: Multimodal Relative Debugging of Failed UI Test Executions *Feb 2026 - June 2026*

- Developed an **automated debugging** pipeline that compares pass/fail UI test traces (actions, logs, screenshots) to rank steps most likely causing a regression and generate technical and plain-language root-cause reports. Combined heuristic (text, pixel diff, visual causal), LLM and VLM scores in a unified fusion architecture (**81%** Hit@1 hybrid, **28%** improvement over heuristic baseline).

- **Github**: https://github.com/MAKhan22/TraceLens-Automated-Debugging

Publications and Preprints

Camera Ready / Accepted

- Khan, M. A. SU NLP 29 at SemEval-2026 Task 5: “DynaOrd - Hybrid Dynamic Ordinal Regression with LoRA-Fine-Tuned DeBERTa-v3.” SemEval 2026 (co-located with ACL 2026).
- M. A. Khan and M. S. Salem, "Anatomy-Aware Routing for Robust Medical Image Segmentation: A Failure-Detectable Alternative to Mixed Models,"IEEE SIU 2026.

Projects

Spatial Analysis of Political Disorder in Kenya (1997–2024) *Oct 2025 - Jan 2026*

- Analyzed 16,225 **ACLED** political violence and protest events (1997–2024) and demonstrated an accessibility gradient in political disorder, with riots concentrated near major roads (median ≈0.35 km) and battles occurring in remote regions, and quantified systematic coastal vs. inland differences in event frequency and severity.
- **Github:** https://github.com/Spatial-Data-Science-494/Kenya-Political-Violence-2022

Github Developer Network Analysis *Jan 2025 - May 2025*

- Analyzed 1.9M **GitHub** users and 3.9M follower relationships using cuGraph and NetworkX and investigated whether programming language diversity correlates with social network centrality using PageRank, betweenness, and community metrics.
- **Github:** https://github.com/GraphVader/fictional-octo-waddle

Understanding a Person’s Ethnicity Using a Photograph *Oct 2023 - Jan 2024*

- Built an image classification **CNN**-based **CV** model with AI to predict ethnicity from photographs using a dataset of individuals from the GCC, Egypt, and Levant regions.

Work Experience

Eastern Al Maha Company (EMCO), *Part-Time Full-Stack Developer* | Kuwait City, Kuwait *Nov 2025 - Jan 2026*

- Lead further refinement of the MACS 3 auditing system and delivered a more feature-rich, powerful and intuitive solution based on field deployment feedback.
- Collaborated in a 2 person team to enhance all 3 parts of the system to ensure long-term maintainability of the company’s nationwide auditing infrastructure. Responsible for tailoring the microservices, database and mobile app to company needs.

Eastern Al Maha Company (EMCO), *Software Developer Intern* | Kuwait City, Kuwait *Jun 2025 - Aug 2025*

- Developed “MACS 3”: a 3 component auditing system combining planogram compliance **computer vision model**, mobile task management **app**, and web-based admin **dashboard** to automate FMCG distribution operations of the company across Kuwait.
- Improved compliance accuracy by **67%** within 10 days, with **80%** of non-compliance flags now originating from the mobile app.
- Responsible for developing Backend Kubernetes microservices task, photos, override requests and reports along with PostgreSQL database structure and automated database management system. Also responsible for developing Flutter Mobile App.

Skills

Natural Languages: English – Professional Proficiency | **Hindi** – Native Proficiency | **Urdu** – Native Proficiency

Computing Languages: C++, Python, R, Java, Prolog, Verilog, SQL, TypeScript, JavaScript, HTML & CSS

Technical: MySQL, PostgreSQL, MongoDB, Firebase, CUDA, Numba, Pandas, PyTorch, Scikit-learn, Numpy, NetworkX, cuGraph, UNet++, Node.js, Express.js, React.js, React Native, Git, Github, Jest, Kubernetes, Docker, Skaffold, Android Studio, Ollama, Postman, Scrum, Jira, LaTeX

Soft: Hardworking, Creativity, Critical thinking, Time management, Teamwork, Communication, Problem solving, Empathy

Certificates

NVIDIA - Fundamentals of Accelerated Computing with CUDA Python *May 2025*

NVIDIA - Applications of AI for Predictive Maintenance *Feb 2025*

NVIDIA - Generative AI with Diffusion Models *Mar 2024*

NVIDIA - Building Transformer-Based Natural Language Processing Applications *Mar 2024*

NVIDIA - Fundamentals of Accelerated Computing with CUDA C/C++ *Mar 2024*

IBM - Data Analysis with Python *Oct 2022*

MITx X Series Program - Computational Thinking using Python *Dec 2021*

HarvardX CS50AI - CS50’s Introduction to Artificial Intelligence with Python *Aug 2021*